INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Case 1: Android Forensics on a Google Pixel 3 (Android 10)

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Course Title: Mobile and IoT Forensics Case Study

Course Code: SBT-DF205

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This report presents an offline examination of the supplied training evidence for Case 1 (SBT-DF205-C1): a 5.25 GB file-system extraction of a company-issued Google Pixel 3 running Android 10 (Joshua Hickman dataset, Digital Corpora), examined for Harbour Operations Ltd after suspected unauthorised communication and sharing of information. The scenario is training data. The archive was hashed before extraction, and each finding below rests on a cited file path, record or screenshot. Dates are historical dates from the dataset.

Principal findings

1. **Evidence integrity.** Pixel 3.zip (5,247,820,897 bytes) has SHA-256 `ca6918ef...8aa00a`, which equals the hash given in the course slides. The first extraction attempt failed for lack of disk space and is recorded below.
2. **Device and profiles.** Google Pixel 3 (blueline), Android 10, build QQ1A.200105.003, patch 2020-01-01, rooted before imaging. Two users exist: 0 ("This Is") and 11 ("User 2"). The account thisisdfir@gmail.com is tied to user 0 by three sources; thisisdfirtwo@gmail.com is not tied to a profile.
3. **Connectivity.** Bluetooth name and address are stored; the Wi-Fi store has an empty network list and one MAC-address entry; the carrier is Google Fi. These are saved settings, not proof of a connection at a given time.
4. **Messages.** The standard mmssms.db holds at least 11 received SMS, all automatic verification messages from LINE, Telegram, Signal, imo, Skout, WhatsApp, Microsoft, TikTok and Facebook, received between 18:20:45 and 18:50:24 UTC on 2020-01-29. Three were matched to Google Messages. The user_de copy of mmssms.db is a different file whose row-count queries returned 0.
5. **WhatsApp and calls.** WhatsApp holds a three-message exchange with 19195790479 on 2020-02-08 (20:55 to 20:58 UTC). The call log holds 14 entries, including four calls with the same number on 2020-02-14 (01:29 to 14:41 UTC). The messages contain no sign of information being shared.
6. **Media.** Two photographs carry GPS tags about 570 m apart (taken 2020-02-09 and 2020-02-13, time zone unresolved). No Google Maps entry was found by the search run.

Confidence. High for what each record shows, because every value can be checked by path and field. Low for any inference about intent: the records examined contain nothing that supports or contradicts the allegation, so the conclusion is **unresolved**. No person-level attribution is made because the evidence does not support one.

EVIDENCE ACQUISITION AND INTEGRITY

Evidence inventory

Table 1 lists the supplied original and the working tree. The SHA-256 was recorded before extraction (Figure 1).

ID	Item	Bytes	SHA-256	Role and location
EVID-01	Pixel 3.zip	5,247,820,897	`ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a`	Original archive, ~/Downloads/pixel3/. MD5 `9cc37ebbbc4e918ee5427de1fe1deecc`.

Table 1. Evidence inventory. Source: Digital Corpora, Non-Cellebrite Extraction, "Pixel 3.zip" (link in the case brief).

Preservation steps and precautions

- **Identification and hashing:** the size came from `ls -l` (file dated 27 Oct 14:06, VM local time) and the hashes from `hashdeep` before extraction. The SHA-256 matches the value on slide 6 of the first course deck (Figure 1).
- **Read-only work:** after extraction only read commands (cat, grep, tree, ls, sha256sum, sqlite3 SELECT queries, exiftool) were used. No live account was accessed and nothing was rooted or reset. DB Browser opened telephony.db in place.
- **Sidecars:** no -wal or -shm files exist in user_de/0/com.android.providers.telephony/databases (Figure 3).
- **Secrets:** no passwords, tokens or keys are reported, verification codes in SMS bodies are omitted, and the Wi-Fi integrity block is redacted (Figure 7).
- **Tools:** hashdeep, unzip, tree, grep, sqlite3, DB Browser for SQLite and ExifTool.

Handling error and verification

The first `unzip -q` stopped with "No space left on device" and left a partial tree (Figure 1).
The archive itself was not altered by the failed extraction.

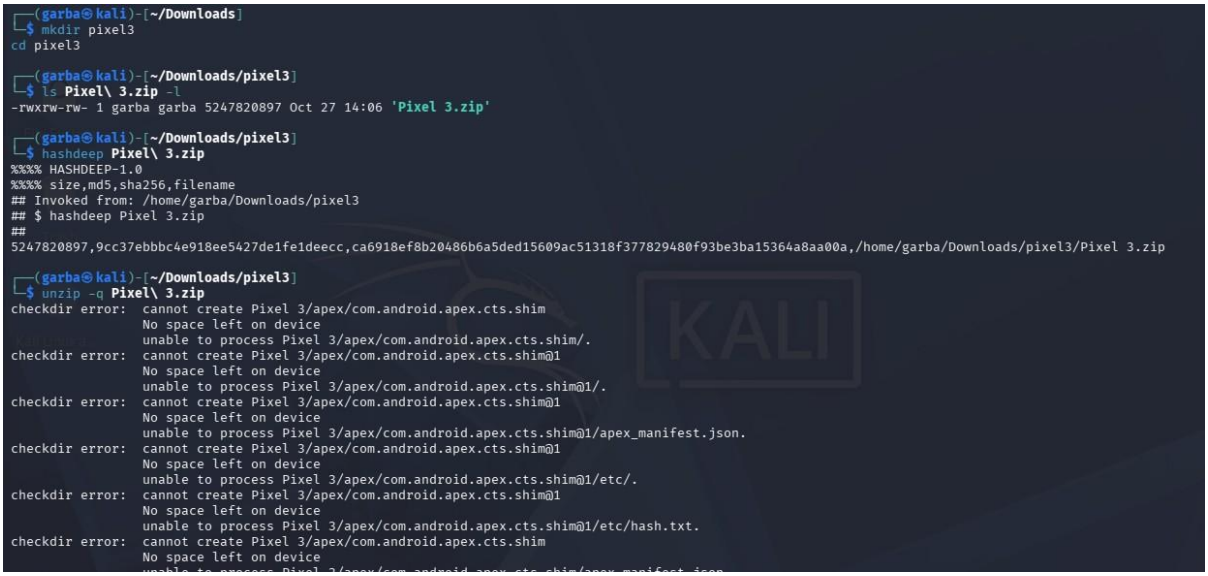


Figure 1: Archive size, SHA-256 and MD5 (hashdeep), and the first unzip attempt failing with "No space left on device" (E01).

METHOD

The examination followed the order in the assessment brief: preserve the evidence, identify the device, review profiles, settings and connectivity, examine communications, applications and media, build a UTC timeline and conclude. Every finding cites a source path under Pixel 3/ and the stored key or field. SQL was run with `sqlite3` against the extracted files, and the queries are listed in Queries.sql.

Time handling

Android XML settings, the Telephony and Google Messages databases, WhatsApp and the call log store 13-digit millisecond Unix times, so each was divided by 1000 before conversion with `datetime(x/1000,'unixepoch')`; `ro.product.build.date.utc` is in seconds. Raw values are kept beside the UTC values. UTC-05:00 is quoted for context because the dataset documentation uses it. ExifTool times are camera-local strings with no offset in the output captured, so photo times are kept explicitly **unresolved** and are not converted. A value with no usable epoch, such as user 0 “created=0”, is also unresolved.

FINDINGS

Summary of answers

Task	Finding	Evidence
1	Archive hashed before extraction and matches the course hash; one extraction error recorded; final hash recheck pending.	E01
2	Pixel 3, Android 10, build QQ1A.200105.003, patch 2020-01-01. Users 0 and 11. thisisdfir@gmail.com tied to user 0 by three sources.	E02 to E06, E10
3	Bluetooth name and address stored; Wi-Fi network list empty with one MAC-map entry; carrier Google Fi.	E07, E08
4	Two mmssms.db copies differ by hash; at least 11 received SMS in the standard copy; 14 call-log rows. MMS, contacts link, read flags and calendar pending.	E11, E12, E15
5	Three SMS match between bugle_db and mmssms.db by text and raw time. Google Dialer comparison pending.	E11, E13
6	WhatsApp exchange of three messages with 19195790479; account identity pending. TextNow and Kik: no records extracted.	E09, E14, E16

7	Two geotagged photographs and one video without EXIF time; no Maps entry found; hashes and Google Photos pending.	E17, E18
8	Sixteen-event timeline across five families, four correlations; conclusion unresolved.	E04 to E17

Task 1. How was the evidence preserved?

Finding. The archive was identified by name, size and SHA-256 before extraction, and the SHA-256 equals the trusted value from the course slides, so no discrepancy exists (Table 1, Figure 1). **Evidence.** E01. **Limit.** A matching hash shows the file equals the one supplied; it does not show the package was complete before it reached the examiner.

Task 2. Which device is it, and who are the users?

Attribute	Property key	Value	Source (under Pixel 3/)
Manufacturer	`ro.product.product.manufacturer`	Google	product/build.prop
Model	`ro.product.product.model`	Pixel 3	product/build.prop
Android version	`ro.product.build.version.release`	10 (SDK 29)	product/build.prop
Build ID	`ro.product.build.id`	QQ1A.200105.003	product/build.prop
Security patch	`ro.build.version.security_patch`	2020-01-01	system/build.prop
Hardware serial	“Serial Number” line	8CEX1N716	mnt/vendor/persist/sensors/tmd2725_als_factory.txt

Table 2. Device properties with their keys (E02, E03, E10). The product-partition keys are shown; the system partition holds its own copies.



Figure 2: product/build.prop: manufacturer, model, Android version, build ID and fingerprint (E02).



Figure 3: system/build.prop security patch level, and the telephony user_de folder with no -wal or -shm files (E03).

Finding	Source (under Pixel 3/)	Detail
Two users	data/system/users/userlist.xml	Users 0 and 11. deviceOwnerUserId is -10000, so no device owner is set.
User 0	data/system/users/0.xml	Name “This Is”, serial 0, flags 19, created=“0” (unset), lastLoggedIn 1581695517124.
User 11	data/system/users/11.xml	Name “User 2”, serial 11, flags 16, created 1581644358241, lastLoggedIn 1581684601922.
Account 1	data/data/com.google.android.gm/shared_prefs/Gmail.xml, UnifiedEmail.xml	active-account thisisdfir@gmail.com; display name “This Is DFIR”.
Corroboration	data/data/com.android.vending/cache/logs/	A log folder named for thisisdfir@gmail.com. With the first name “This Is” on user 0, three sources agree. A Facebook SMS (Task 4) also names “ThisIs”.
Account 2	data/system/job/jobs.xml	thisisdfirtwo@gmail.com appears 36 times (thisisdfir@gmail.com 43 times). Not tied to a profile.

Table 3. User profiles and account associations (E04 to E06).

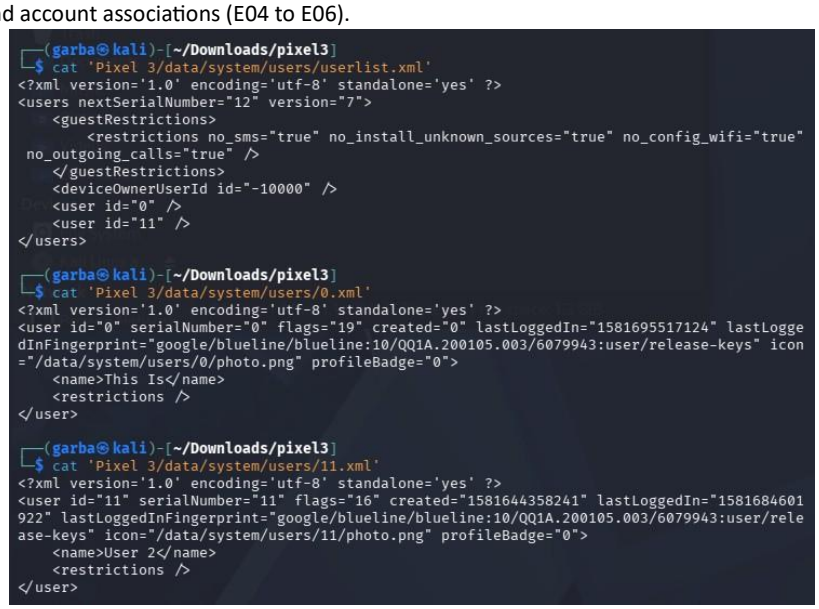


Figure 4: userlist.xml, 0.xml and 11.xml: users, names, creation and last-login values (E04).

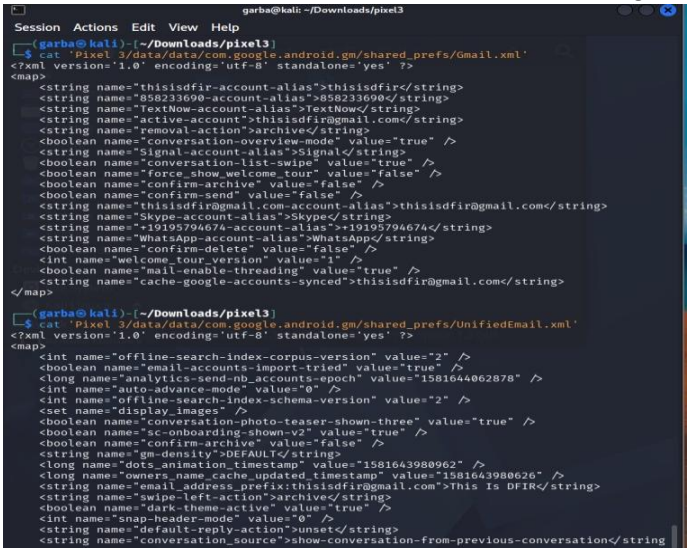


Figure 5: Gmail.xml and UnifiedEmail.xml: active account thisisdfir@gmail.com and the owner name (E05).



Figure 6: Play Store log folders, address counts from jobs.xml, and the system_ce and system_de folders (E06).

Interpretation. The device is a Pixel 3 on Android 10 with two profiles, and the main Google account belongs with user 0. **Limit.** Names, accounts and login times describe device state; they do not show who held the phone at a given moment. Activity of users 0 and 11 is kept separate, and tying the second address to user 11 needs `accounts_ce.db`. The phone was rooted before imaging (Magisk files under sbin/.magisk and the com.topjohnwu.magisk package), so its state may differ from a stock handset.

Task 3. What do the settings and connectivity records show?

Setting or record	Stored key and value	Source (under Pixel 3/)
Bluetooth name	`bluetooth_name` (id 65) = “Pixel 3”	data/system/users/0/settings_secure.xml
Bluetooth address	`bluetooth_address` (id 71) = 7C:D9:5C:AC:A2:CE	data/system/users/0/settings_secure.xml
Wi-Fi networks	NetworkList is empty	data/misc/wifi/WifiConfigStore.xml
Wi-Fi MAC map	Key “CcookiesDcastleR5” WPA_PSK = ae:f6:c7:a8:a5:c2	data/misc/wifi/WifiConfigStore.xml
Carrier	`carrier_name_string` = Google Fi (two carrierconfig files)	data/user_de/0/com.android.phone/files/
Subscriber	`subscriberId` 310260974867669 (mobile template)	data/system/netpolicy.xml

Table 4. Settings and connectivity records (E07, E08).

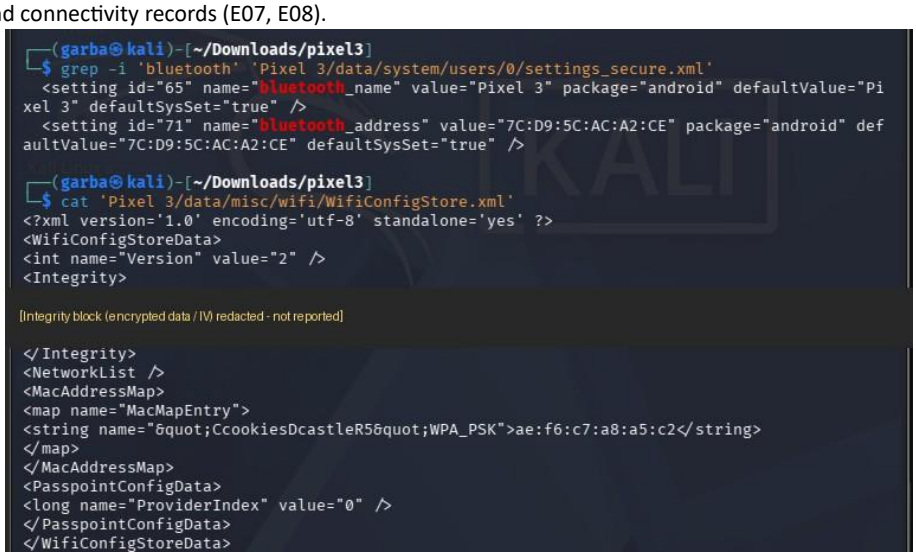


Figure 7: Bluetooth name and address, and WifiConfigStore.xml. The encrypted integrity block is redacted (E07).

Limit. Saved configuration is not a proven connection. The empty list means no saved Wi-Fi network remained in the store, while the MAC-map entry shows Android once held a per-network address for that name. Neither shows when, or whether, the phone joined that network, and no Bluetooth pairing list has been examined [pending].

Task 4. What do the SMS, call and contact records show?

Telephony copies. The two copies of mmssms.db are different files.

Path (under Pixel 3/)	SHA-256	Row counts
data/data/com.android.providers.telephony/databases/mmssms.db	cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342	sms: at least 11 rows (ids 2 to 12 shown). Full counts of sms and pdu
data/user_de/0/com.android.providers.telephony/databases/mmssms.db	d0ce9034defe1f2ba0409f1c86947447cd90f414a4fda5d847804d01ef34c827	Two count(*) queries returned 0 and 0.

Table 5. Telephony Provider copies by path, hash and row count (E11, E12).

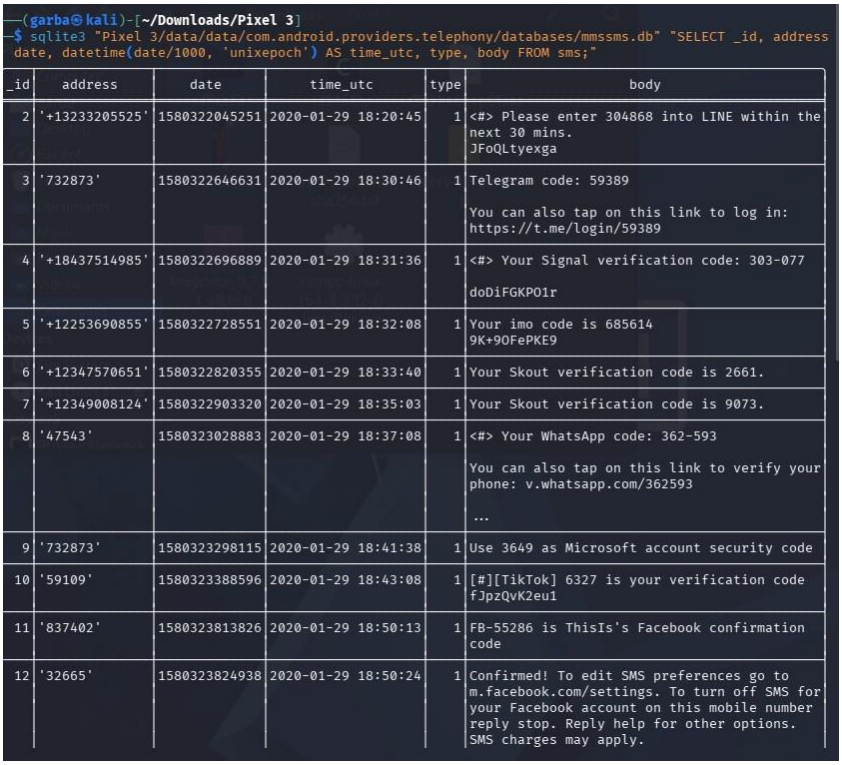


Figure 8: sha256sum of the two mmssms.db copies, and the row-count queries on the user_de copy (E11, E12).

SMS records. Five of the SMS rows are shown below; the rest are in Figure 9. Every SMS shown has type 1, which is a received message, and none was sent from the phone.

ID	Address (as shown)	Raw (ms)	UTC	Dir.	Content
2	'+13233205525'	1580322045251	2020-01-29 18:20:45	In	LINE verification message (code omitted)
3	'732873'	1580322646631	2020-01-29 18:30:46	In	Telegram login code message
4	'+18437514985'	1580322696889	2020-01-29 18:31:36	In	Signal verification code
5	'+12253690855'	1580322728551	2020-01-29 18:32:08	In	imo verification code
8	'47543'	1580323028883	2020-01-29 18:37:08	In	WhatsApp verification code and link

Table 6. SMS records from the standard mmssms.db (E11). Numbers are shown as they appear in the query output, including the single quotes. The read flag was not selected.



```
┌──(garba㉿kali)-[~/Downloads/Pixel 3]
└─$ sqlite3 "Pixel 3/data/data/com.android.providers.telephony/databases/mmssms.db" "SELECT _id, address
date, datetime(date/1000, 'unixepoch') AS time_utc, type, body FROM sms;"
```

_id	address	date	time_utc	type	body
2	'+13233205525'	1580322045251	2020-01-29 18:20:45	1	<#> Please enter 304868 into LINE within the next 30 mins. JFoQLtyexga
3	'732873'	1580322646631	2020-01-29 18:30:46	1	Telegram code: 59389 You can also tap on this link to log in: https://t.me/login/59389
4	'+18437514985'	1580322696889	2020-01-29 18:31:36	1	<#> Your Signal verification code: 303-077 doDiFGKPO1r
5	'+12253690855'	1580322728551	2020-01-29 18:32:08	1	Your imo code is 685614 9K+9OFePKE9
6	'+12347570651'	1580322820355	2020-01-29 18:33:40	1	Your Skout verification code is 2661.
7	'+12349008124'	1580322903320	2020-01-29 18:35:03	1	Your Skout verification code is 9073.
8	'47543'	1580323028883	2020-01-29 18:37:08	1	<#> Your WhatsApp code: 362-593 You can also tap on this link to verify your phone: v.whatsapp.com/362593 …
9	'732873'	1580323298115	2020-01-29 18:41:38	1	Use 3649 as Microsoft account security code
10	'59109'	1580323388596	2020-01-29 18:43:08	1	[#][TikTok] 6327 is your verification code fJpzQvK2eu1
11	'837402'	1580323813826	2020-01-29 18:50:13	1	FB-55286 is ThisIs's Facebook confirmation code
12	'32665'	1580323824938	2020-01-29 18:50:24	1	Confirmed! To edit SMS preferences go to m.facebook.com/settings. To turn off SMS for your Facebook account on this mobile number reply stop. Reply help for other options. SMS charges may apply.

Figure 9: SMS records from the standard mmssms.db: IDs, addresses, raw times, UTC times, type and body (E11).

The 11 rows shown are verification messages from LINE, Telegram, Signal, imo, Skout (two), WhatsApp, Microsoft, TikTok and Facebook, received within 30 minutes. They show services being set up on the number, not any conversation. All are inbound, so no two-way SMS conversation is present in these rows

Call log. calllog.db holds 14 rows (ids 1 to 14). Type 1 is incoming, 2 outgoing, 3 missed and 4 voicemail (Android call-log constants). Ids 1 to 5 store no number.

ID	Number (as shown)	Raw (ms)	UTC	Dur. (s)	Type
6	'+332629243087'	1581192896305	2020-02-08 20:14:56	0	3 missed
7	'+19195121037'	1581521849939	2020-02-12 15:37:29	0	3 missed
8	'+19195121037'	1581521901000	2020-02-12 15:38:21	3	4 voicemail
11	'9195790479'	1581643741051	2020-02-14 01:29:01	11	2 outgoing
12	'+19195790479'	1581643824061	2020-02-14 01:30:24	65	1 incoming
13	'9195790479'	1581685557104	2020-02-14 13:05:57	86	2 outgoing
14	'9195790479'	1581691278351	2020-02-14 14:41:18	91	1 incoming

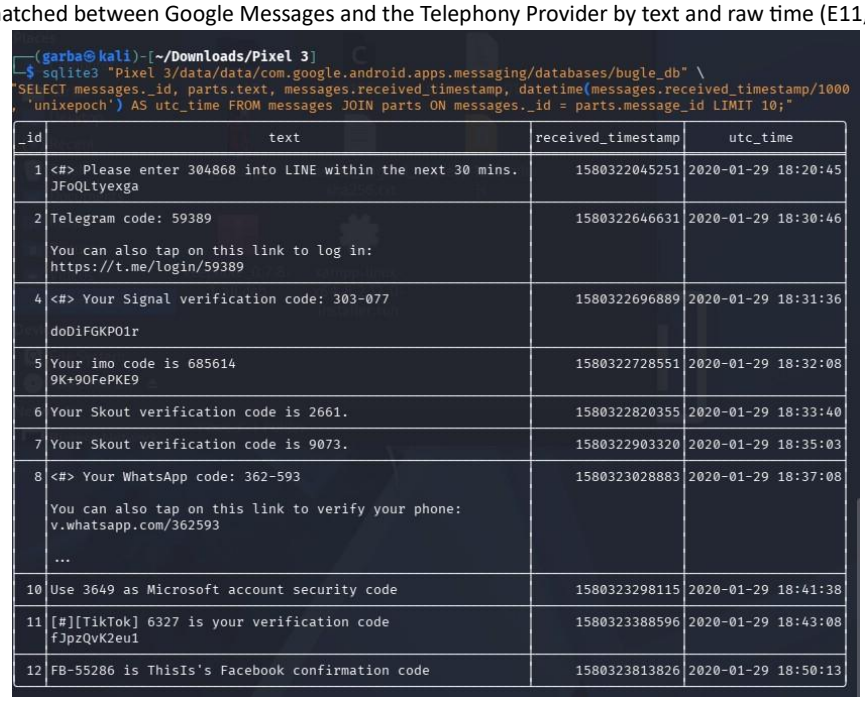
Table 7. Selected call-log rows (E15). The same subscriber appears as 9195790479 and +19195790479.

Contacts and calendar. Linking two numbers to contacts in contacts2.db, and the calendar table names and row counts, have no captured records yet. **Limit.** A call-log row records the number, time, duration and type; it does not record which profile was in use or what was said.

Task 5. Do Google Messages and the Dialer agree with the AOSP records?

Message	mmssms.db id	bugle_db id	Raw received time (both)	UTC
LINE verification	2	1	1580322045251	2020-01-29 18:20:45
Telegram login code	3	2	1580322646631	2020-01-29 18:30:46
Signal verification	4	4	1580322696889	2020-01-29 18:31:36

Table 8. Three SMS matched between Google Messages and the Telephony Provider by text and raw time (E11, E13).



```
┌──(garba㉿kali)-[~/Downloads/Pixel 3]
└─$ sqlite3 "Pixel 3/data/data/com.google.android.apps.messaging/databases/bugle_db" \
"SELECT messages._id, parts.text, messages.received_timestamp, datetime(messages.received_timestamp/1000
, 'unixepoch') AS utc_time FROM messages JOIN parts ON messages._id = parts.message_id LIMIT 10;"
```

_id	text	received_timestamp	utc_time
1	<#> Please enter 304868 into LINE within the next 30 mins. JFoQLtyexga	1580322045251	2020-01-29 18:20:45
2	Telegram code: 59389 You can also tap on this link to log in: https://t.me/login/59389	1580322646631	2020-01-29 18:30:46
4	<#> Your Signal verification code: 303-077 doDiFGKPO1r	1580322696889	2020-01-29 18:31:36
5	Your imo code is 685614 9K+9OFePKE9	1580322728551	2020-01-29 18:32:08
6	Your Skout verification code is 2661.	1580322820355	2020-01-29 18:33:40
7	Your Skout verification code is 9073.	1580322903320	2020-01-29 18:35:03
8	<#> Your WhatsApp code: 362-593 You can also tap on this link to verify your phone: v.whatsapp.com/362593 ...	1580323028883	2020-01-29 18:37:08
10	Use 3649 as Microsoft account security code	1580323298115	2020-01-29 18:41:38
11	[#][TikTok] 6327 is your verification code fJpzQvK2eu1	1580323388596	2020-01-29 18:43:08
12	FB-55286 is ThisIs's Facebook confirmation code	1580323813826	2020-01-29 18:50:13

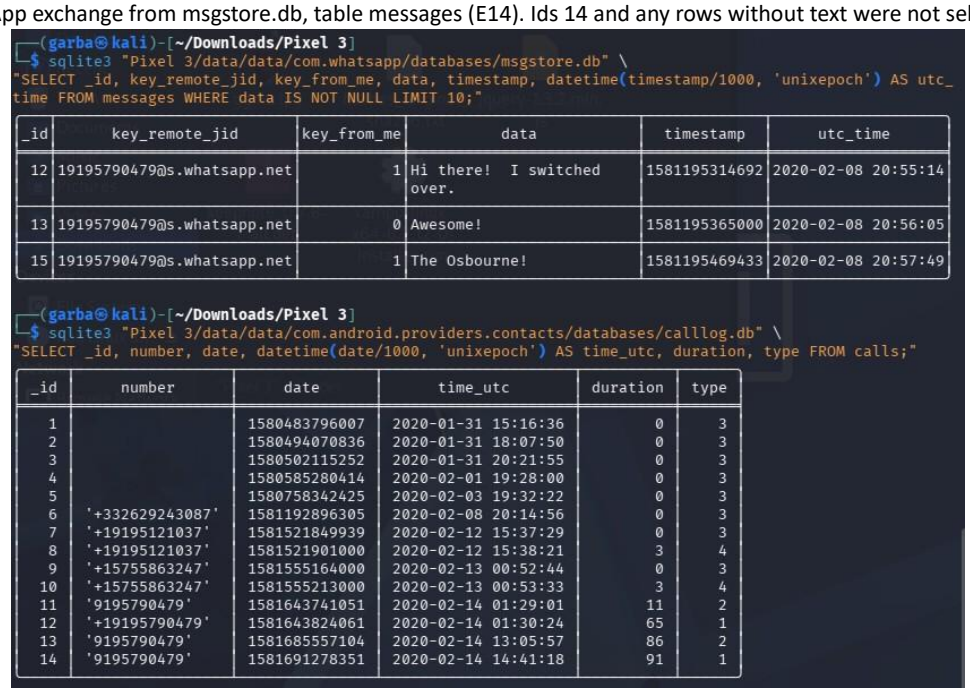
Figure 10: Google Messages bugle_db: messages joined to parts, with raw and UTC received times (E13).

Finding. The text and raw millisecond time are identical in both databases for these three messages, and the other rows shown in Figure 10 match too. The record IDs differ (for example 2 against 1, and 3 against 2), so IDs cannot be used to link the two stores; text and time were used. The bugle_db query selected no address column, so the sender was not compared

Task 6. What do WhatsApp, TextNow and Kik show?

ID	Participant (key_remote_jid)	Dir. (key_from_me)	Content	Raw (ms)	UTC
12	19195790479@s.whatsapp.net	1 (sent)	Hi there! I switched over.	1581195314692	2020-02-08 20:55:14
13	19195790479@s.whatsapp.net	0 (received)	Awesome!	1581195365000	2020-02-08 20:56:05
15	19195790479@s.whatsapp.net	1 (sent)	The Osbourne!	1581195469433	2020-02-08 20:57:49

Table 9. WhatsApp exchange from msgstore.db, table messages (E14). Ids 14 and any rows without text were not selected.



```
┌──(garba㉿kali)-[~/Downloads/Pixel 3]
└─$ sqlite3 "Pixel 3/data/data/com.whatsapp/databases/msgstore.db" \
"SELECT _id, key_remote_jid, key_from_me, data, timestamp, datetime(timestamp/1000, 'unixepoch') AS utc_
time FROM messages WHERE data IS NOT NULL LIMIT 10;"
```

_id	key_remote_jid	key_from_me	data	timestamp	utc_time
12	19195790479@s.whatsapp.net	1	Hi there! I switched over.	1581195314692	2020-02-08 20:55:14
13	19195790479@s.whatsapp.net	0	Awesome!	1581195365000	2020-02-08 20:56:05
15	19195790479@s.whatsapp.net	1	The Osbourne!	1581195469433	2020-02-08 20:57:49

```
┌──(garba㉿kali)-[~/Downloads/Pixel 3]
└─$ sqlite3 "Pixel 3/data/data/com.android.providers.contacts/databases/calllog.db" \
"SELECT _id, number, date, datetime(date/1000, 'unixepoch') AS time_utc, duration, type FROM calls;"
```

_id	number	date	time_utc	duration	type
1		1580483796007	2020-01-31 15:16:36	0	3
2		1580494070836	2020-01-31 18:07:50	0	3
3		1580502115252	2020-01-31 20:21:55	0	3
4		1580585280414	2020-02-01 19:28:00	0	3
5		1580758342425	2020-02-03 19:32:22	0	3
6	'+332629243087'	1581192896305	2020-02-08 20:14:56	0	3
7	'+19195121037'	1581521849939	2020-02-12 15:37:29	0	3
8	'+19195121037'	1581521901000	2020-02-12 15:38:21	3	4
9	'+15755863247'	1581555164000	2020-02-13 00:52:44	0	3
10	'+15755863247'	1581555213000	2020-02-13 00:53:33	3	4
11	'9195790479'	1581643741051	2020-02-14 01:29:01	11	2
12	'+19195790479'	1581643824061	2020-02-14 01:30:24	65	1
13	'9195790479'	1581685557104	2020-02-14 13:05:57	86	2
14	'9195790479'	1581691278351	2020-02-14 14:41:18	91	1

Figure 11: WhatsApp msgstore.db messages (top) and the AOSP call log, calllog.db (bottom) (E14, E15).

WhatsApp. The schema in this evidence uses a messages table with key_remote_jid, key_from_me, data and a millisecond timestamp. Three messages with one participant form a short exchange: the device sent a greeting, the participant answered, and the device replied. The same number, 9195790479, appears in four calls on 2020-02-14 (Table 7), which corroborates the participant from a second source. The WhatsApp verification SMS (Table 6, id 8) arrived on 2020-01-29, ten days earlier. The grep for “phone” in com.whatsapp_preferences.xml returned no output, so the account identity is not yet extracted [try grep -iE "jid|registration|phone" on the preferences file, and read wa.db], and the meaning of message status values is not validated. The contents are short and personal in tone and show no sharing of information.

TextNow and Kik. TextNow is registered as an account type for user 0 (AccountAuthenticator.xml, uid 10239) and appears as an alias in Gmail.xml, which is an account association (E05, E09). The queries on textnow.db (table messages) and kikDatabase.db (table bodyTable) printed nothing with errors hidden, and .tables on both printed nothing (Figure 12). No record was extracted, so this is an inconclusive result, not proof that no messages exist: the database file names or paths may differ

Task 7. What do the Maps and photograph records show?

File (under Pixel 3/data/media/0/DCIM/Camera/)	DateTimeOriginal	Offset	GPS (as printed)	Decimal degrees
IMG_20200209_132353.jpg	2020:02:09 13:23:53	Unresolved	35 deg 39' 39.53" N, 78 deg 52' 47.63" W	35.660981, -78.879897
IMG_20200213_195030.jpg	2020:02:13 19:50:31	Unresolved	35 deg 39' 31.92" N, 78 deg 52' 27.18" W	35.658867, -78.874217
2020-02-09-125547759.mp4	none returned	n/a	none returned	n/a

Table 10. ExifTool output for three media files (E17). SHA-256 values [add sha256sum for each file].

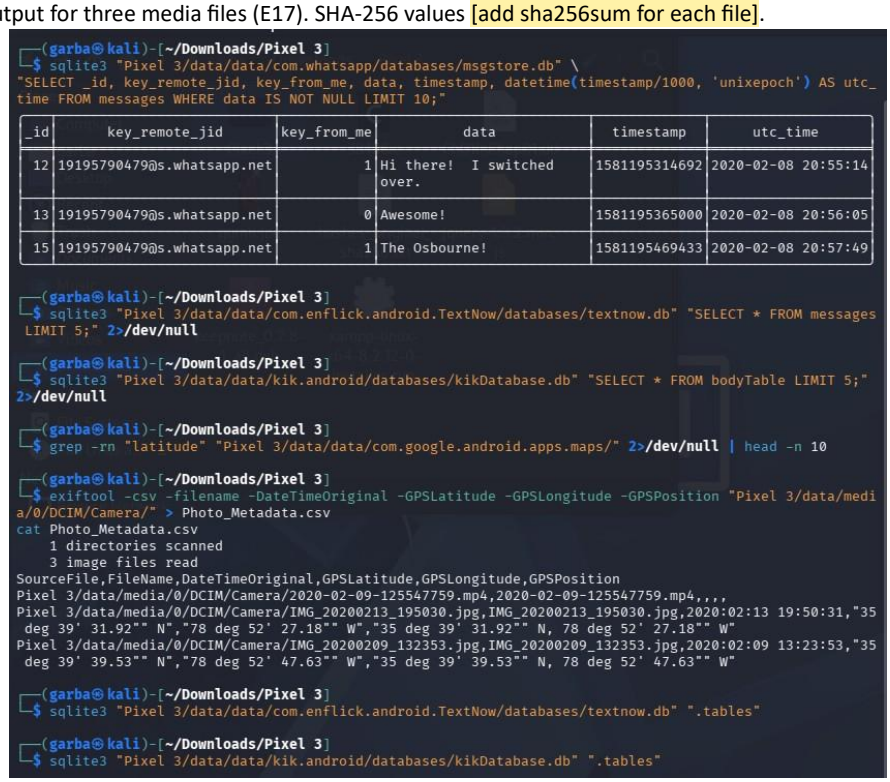


Figure 12: WhatsApp, TextNow and Kik queries, the Maps search, the ExifTool CSV for DCIM/Camera, and TextNow and Kik .tables (E14, E16 to E18).

Photographs. ExifTool read three files in the Camera folder. Two JPEGs carry a capture time and GPS tags; the video returned neither. The capture times print without an offset, so they stay **unresolved** and are not converted. The two coordinates lie about 570 m apart by straight-line distance, which is my own calculation from the decimal values. **Maps.** A recursive search for “latitude” in the com.google.android.apps.maps folder returned no output, so no Maps entry was

found by that method. [list the Maps databases with find, identify two location entries and say whether each is a search, saved place or route; examine Google Photos databases] **Comparison limit.** A Maps-to-photo comparison could not be made without a Maps entry. Even then, a geotag shows where the device reported being when the file was made, not who carried it.

TIMELINE AND CORRELATION

All times are UTC, taken from raw values shown in the tables above. Rows are distinct events from five families: SMS, WhatsApp, calls, photographs and profile or account files. The same SMS in Google Messages counts once.

#	Time (UTC)	Family and source	Event	Evidence
1	2020-01-29 18:20:45	SMS, mmssms.db id 2	First of 11 received verification SMS (LINE).	E11
2	2020-01-29 18:37:08	SMS, mmssms.db id 8	WhatsApp verification SMS received.	E11
3	2020-01-29 18:50:24	SMS, mmssms.db id 12	Last SMS shown (Facebook confirmation).	E11
4	2020-02-08 20:55:14	WhatsApp, messages id 12	Message sent to 19195790479.	E14
5	2020-02-08 20:56:05	WhatsApp, messages id 13	Reply received from 19195790479.	E14
6	2020-02-08 20:57:49	WhatsApp, messages id 15	Second message sent to 19195790479.	E14
7	2020-02-09 13:23:53 (unresolved)	Photo, IMG_20200209_132353.jpg	Photograph taken; GPS 35.660981, -78.879897.	E17
8	2020-02-13 19:50:31 (unresolved)	Photo, IMG_20200213_195030.jpg	Photograph taken; GPS 35.658867, -78.874217.	E17
9	2020-02-14 01:29:01	Call, calllog.db id 11	Outgoing call to 9195790479, 11 s.	E15
10	2020-02-14 01:30:24	Call, calllog.db id 12	Incoming call from +19195790479, 65 s.	E15
11	2020-02-14 01:33:00	Gmail, UnifiedEmail.xml	Owner-name cache written for thisisdfir@gmail.com.	E05
12	2020-02-14 01:39:18	Profile, users/11.xml	User 11 created.	E04
#	Time (UTC)	Family and source	Event	Evidence
13	2020-02-14 12:50:01	Profile, users/11.xml	User 11 last logged in.	E04
14	2020-02-14 13:05:57	Call, calllog.db id 13	Outgoing call to 9195790479, 86 s.	E15
15	2020-02-14 14:41:18	Call, calllog.db id 14	Incoming call from 9195790479, 91 s.	E15
16	2020-02-14 15:51:57	Profile, users/0.xml	User 0 last logged in.	E04

Table 11. Chronological timeline. Raw values, field names and units are in Timeline.csv. UTC-05:00 context: subtract five hours.

Correlation across artefacts

- **Same message in two stores.** LINE, Telegram and Signal messages appear in both mmssms.db and bugle_db with identical text and raw time (Table 8). This corroborates the SMS records and counts as one event each.
- **Same number in two apps.** 19195790479 is the WhatsApp participant on 2020-02-08 (events 4 to 6) and the number in four calls on 2020-02-14 (events 9, 10, 14 and 15), written as 9195790479 or +19195790479. This links the two records to one subscriber number, but nobody is named because the contacts link is pending.
- **Calls against profile events.** The first two calls (01:29 and 01:30) fall within ten minutes before the Gmail owner-name cache write and the creation of user 11 (events 11 and 12). The later calls at 13:05:57 and 14:41:18 lie between user 11's last login (12:50:01) and user 0's (15:51:57). This is a timing pattern only: the call log does not record which profile was in use.

- **Service setup then use.** The WhatsApp verification SMS (event 2) precedes the WhatsApp messages by ten days, which is consistent with the account being registered on this number, not proof of who registered it.
- **Does not establish.** Photo times have no offset, so no photo can be placed against a call or message; the contents seen show no sharing of information.

CONCLUSION AND LIMITATIONS

Provisional conclusion: unresolved. The records examined show a rooted Pixel 3 with two profiles, a Google account tied to user 0, messaging accounts set up by SMS verification on 2020-01-29, a short WhatsApp exchange with 19195790479 on 2020-02-08, calls with the same number on 2020-02-14, and two geotagged photographs. None of these records contains or describes unauthorised communication or sharing of information, so the records neither support nor contradict the allegation. Several sources that could decide the matter (TextNow or Kik, Google Maps, Google Photos, contacts, MMS) have not yet yielded records.

Limitations

1. **Profile and account are not a person.** Shared use of the phone, a shared account or an unattended device cannot be excluded.
2. **Rooted, logical extraction.** The dataset was copied from a rooted device and is not a physical image, so some state may differ from a stock handset.
3. **Saved configuration is not a connection,** and geotags and saved places do not prove a person visited a place.
4. **Presence is not use.** App folders and registered account types show installation and setup only.
5. **Timestamps.** XML values are job, cache and profile events, not user actions. Photo times have no offset and stay unresolved. User 0 “created” is unset.
6. **Incomplete examination.** The items in yellow have no records yet, the screenshots show only part of each table, and the hash recheck and tool versions are still to be recorded.

Alternative interpretations and further evidence

1. The contact 19195790479 may be a routine personal or business contact, so the WhatsApp messages and calls would be ordinary. **2.** The verification messages may show a person setting up several messaging services, which could be preparation for private communication or simply a new phone; the records cannot say which. **3.** Activity may belong to profile 11 or another person sharing the phone. Deciding between these needs the identity of 19195790479 from contacts2.db, the TextNow or Kik records, WhatsApp media and the account identity, Google Maps and Photos databases, photo hashes with offsets, `accounts_ce.db`, lock-screen and biometric records, and carrier or cloud-account records, which lie outside this activity.

EVIDENCE REGISTER (SUMMARY)

ID	Source path (under Pixel 3/)	Fig.
E01	Pixel 3.zip (original archive)	1
E02	product/build.prop	2
E03	system/build.prop; data/user_de/0/com.android.providers.telephony/ (folder listing)	3
E04	data/system/users/userlist.xml, 0.xml, 11.xml	4
E05	data/data/com.google.android.gm/shared_prefs/Gmail.xml, UnifiedEmail.xml	5
E06	data/data/com.android.vending/cache/logs/; data/system/job/jobs.xml; data/system_ce/0, system_de/0	6
E07	data/system/users/0/settings_secure.xml; data/misc/wifi/WifiConfigStore.xml	7
E08	data/user_de/0/com.android.phone/files/carrierconfig-*.xml (two); data/system/netpolicy.xml	-

E09	data/system/users/0/registered_services/android.accounts.AccountAuthenticator.xml	-
E10	mnt/vendor/persist/sensors/tmd2725_als_factory.txt (hardware serial)	-
E11	data/data/com.android.providers.telephony/databases/mmssms.db	8, 9
E12	data/user_de/0/com.android.providers.telephony/databases/mmssms.db	8
E13	data/data/com.google.android.apps.messaging/databases/bugle_db	10
E14	data/data/com.whatsapp/databases/msgstore.db	11, 12
E15	data/data/com.android.providers.contacts/databases/calllog.db	11
E16	data/data/com.enflick.android.TextNow/databases/ and data/data/kik.android/databases/ (no records returned)	12
E17	data/media/0/DCIM/Camera/ (two JPEG, one MP4)	12
E18	data/data/com.google.android.apps.maps/ (search for “latitude”, no output)	12

Table 12. Evidence IDs. File sizes and SHA-256 values are in Evidence_Register.csv.

ACADEMIC INTEGRITY AND AUTHORIZATION STATEMENT

I confirm that I completed this digital forensics case study in an authorized virtual environment using the supplied scenario dataset. The commands, record citations, evidence-log entries, cryptographic hashes and observations in this report are my own work and are accurately documented, including the extraction error described above. Course teaching materials (Android slides, decks 1 to 3) were used as guidance only. Dataset: Joshua Hickman, Android 10 Pixel 3, Digital Corpora.

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